

Riemannian Manifolds

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Abstract

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1 Riemannian Metrics

1.1 The Riemannian Metric

Definition 1.1.1 (Riemannian Metric)

Let M be a smooth manifold. A Riemannian metric on M is a smooth section $g \in \Gamma(M, (TM \otimes TM)^*)$ such that $g_p : T_p(M) \times T_p(M) \rightarrow \mathbb{R}$ is an inner product for all $p \in M$.

Equivalently, a Riemannian metric is an assignment $p \mapsto g_p$ such that for any smooth sections $s, t \in \Gamma(M, TM)$, the map $p \mapsto g_p(s(p), t(p))$ is smooth.

Definition 1.1.2 (Riemannian Manifold) A Riemannian manifold (M, g) is a manifold M together with a Riemannian metric g on M .

To describe the Riemannian metric in local coordinates, we need some simple algebra. Let V be a finite dimensional vector space over a field k . Let $\{e_1, \dots, e_n\}$ be a basis for V . Let $\{e^1, \dots, e^n\}$ be the dual basis for V^* . Then we may establish a canonical isomorphism

$$(V \otimes_k V)^* \cong V^* \otimes_k V^*$$

given by $f(-, -) \mapsto \sum_{i,j=1}^n f(e_i \otimes e_j) e^i \otimes e^j$. Its converse is given by $\varphi \otimes \psi \mapsto ((v \otimes w) \mapsto \varphi(v)\psi(w))$.

Then isomorphism carries over to vector bundles, and so we have an isomorphism $(TM \otimes TM)^* \cong TM^* \otimes TM^*$. This is how the Riemannian metric is a $(0, 2)$ -tensor field.

Proposition 1.1.3

Let (M, g) be a Riemannian manifold. Let $p \in M$. Let $(U, x^1, \dots, x^n)$ be a chart at p . Then g_p as an element of $T_p(M)^* \otimes T_p(M)^*$ is given locally on the chart by

$$g_p = \sum_{1 \leq i, j \leq n} g_{i,j}(p) (dx^i)|_p \otimes (dx^j)|_p$$

where $g_{i,j} : U \rightarrow \mathbb{R}$ are smooth functions such that $g_{i,j}(p) = g_p \left(\left. \frac{\partial}{\partial x^i} \right|_p, \left. \frac{\partial}{\partial x^j} \right|_p \right)$.

Let $v, w \in T_p(M)$ be tangent vectors. Then locally on the chart $(U, x^1, \dots, x^n)$, we may compute the Riemannian metric by

$$g_p(v, w) = \sum_{1 \leq i, j \leq n} g_{i,j}(p) (dx^i)|_p(v) (dx^j)|_p(w)$$

Theorem 1.1.4

Every smooth manifold admits a Riemannian metric and hence is a Riemannian manifold.

Definition 1.1.5 (Isometries) Let (M, g) and (N, h) be two Riemannian manifolds. We say that (M, g) and (N, h) are isometric if there exists a diffeomorphism $f : M \rightarrow N$ such that

$$h \circ f = g$$

In this case f is said to be an isometry.

Theorem 1.1.6 ((Nash))

Let (M, g) be a Riemannian manifold of dimension n . Then there exists an isometric embedding $f : M \rightarrow \mathbb{R}^{(n+2)(n+3)/2}$ with the standard Euclidean metric.

Definition 1.1.7 (Local Isometries) Let (M, g) and (N, h) be two Riemannian manifolds. We say that they are locally isometric if for all $p \in M$, there exists an open neighbourhood $U \subseteq M$ of p and $V \subseteq N$ open and an isometry $f : U \rightarrow V$.

Definition 1.1.8 (Flat Manifolds) Let (M, g) be a Riemannian manifold. We say that (M, g) is flat if it is locally isometric to $\mathbb{R}^n$ with the standard metric.

In general, not every Riemannian manifold is flat. This can be shown once we discuss curvatures and torsions. However, this is true when $n = 1$.

Lemma 1.1.9 Every 1 dimensional Riemannian manifold is flat.

1.2 Lengths and Angles

Definition 1.2.1 (Length of a Tangent Vector) Let (M, g) be a Riemannian manifold. Let $v \in T_p(M)$ be a tangent vector for $p \in M$. Define the length of v to be

$$|v|_g = \sqrt{g_p(v, v)}$$

Definition 1.2.2 (Angle between two Tangent Vectors) Let (M, g) be a Riemannian manifold. Let $p \in M$. For $v, w \in T_p M$ two tangent vectors, define the angle between v and w to be the unique $\theta \in [0, \pi]$ such that

$$\cos(\theta) = \frac{g_p(v, w)}{|v|_g |w|_g}$$

Definition 1.2.3 (Orthogonal Tangent Vectors) Let (M, g) be a Riemannian manifold. Let $p \in M$. We say that two tangent vectors $v, w \in T_p M$ are orthogonal if

$$g_p(v, w) = 0$$

1.3 Musical Isomorphism

Definition 1.3.1 (The Flat Map) Let (M, g) be a Riemannian manifold. Let $p \in M$. For each $X \in T_p M$, define the flat map

$$\flat : T_p M \rightarrow T_p^* M$$

by sending $X \in T_p M$ to the map $X^\flat : T_p M \rightarrow \mathbb{R}$ by $X^\flat(Y) = g_p(X, Y)$.

Theorem 1.3.2 (The Musical Isomorphism) Let (M, g) be a Riemannian manifold. Let $p \in M$. Then the flat map

$$\flat : T_p M \rightarrow T_p^* M$$

is an isomorphism.

Definition 1.3.3 (The Sharp Map) Let (M, g) be a Riemannian manifold. Let $p \in M$. Define the sharp map

$$\sharp : T_p^* M \rightarrow T_p M$$

to be the inverse of the flat map.

Example 1.3.4

Let (M, g) be a Riemannian manifold. Let $p \in U$. Let $(U, x^1, \dots, x^n)$ be a chart at p . Let g be given locally at the chart by

$$g_p = \sum_{1 \leq i, j \leq n} g_{i,j}(p) (dx^i)_p \otimes (dx^j)_p$$

Then the following are true.

- The flat map is given locally by

$$\left(\frac{\partial}{\partial x^i} \Big|_p \right)^b = \sum_{j=1}^n g_{i,j}(p) (dx^j)_p$$

- The sharp map is given locally by

$$((dx^i)_p)^\# = \sum_{j=1}^n g^{i,j}(p) \frac{\partial}{\partial x^j} \Big|_p$$

where $(g^{i,j}(p))_{1 \leq i,j \leq n} = ((g_{i,j}(p))_{1 \leq i,j \leq n})^{-1}$.

1.4 The Induced Metric on Cotangent Bundles

1.5 Riemannian Manifolds as a Metric Space

Definition 1.5.1 (Piecewise Regular Paths)

Let (M, g) be a Riemannian manifold. A piecewise regular path in M consists of smooth functions $\gamma_i : [a_i, b_i] \rightarrow M$ for $1 \leq i \leq k$ such that the following are true.

- For $1 \leq i \leq k-1$, we have $\gamma_i(b_i) = \gamma_{i+1}(a_{i+1})$.
- For $1 \leq i \leq k$, we have $|\gamma'_i(t)| \neq 0$ for all $t \in [a_i, b_i]$.

Definition 1.5.2 (Length of a Curve)

Let (M, g) be a Riemannian manifold. Let $\gamma : [a, b] \rightarrow M$ be a piecewise regular path in M . Define the length of γ to be

$$L(\gamma) = \int_a^b |\gamma'(t)|_g dt$$

Definition 1.5.3 (Riemannian Distance Function)

Let (M, g) be a Riemannian manifold. Define the Riemannian distance function of M to be the function $d_g : M \times M \rightarrow \mathbb{R}$ given by

$$d_g(p, q) = \inf \{ L(\gamma) \mid \gamma : I \rightarrow M \text{ is a piecewise regular path} \}$$

Proposition 1.5.4

Let (M, g) be a Riemannian manifold. Then the following are true.

- d_g is a metric on M .
- The topology of M is given by the metric topology induced by d_g .

2 Connections and Parallel Transports

2.1 Affine Connections

Recall that for a smooth vector bundle $p : E \rightarrow M$, we denote the space of smooth sections on E by $\Gamma(E)$. Moreover, $\mathfrak{X}(M)$ is the space of smooth sections on the tangent bundle TM .

Definition 2.1.1 (Connections) Let M be a smooth manifold. Let $p : E \rightarrow M$ be a smooth vector bundle. A connection on p is a map

$$\nabla : \mathfrak{X}(M) \times \Gamma(E) \rightarrow \Gamma(E)$$

where we denote $\nabla(V, T)$ by $\nabla_V(T)$, such that the following are true.

- $C^\infty(M)$ -linearity in first variable: For each $T \in \Gamma(E)$, the map $V \mapsto \nabla_V(T)$ is $C^\infty(M)$ -linear. This means that

$$\nabla_{fV+gW}(T) = f\nabla_V(T) + g\nabla_W(T)$$

for $V, W \in \mathfrak{X}(M)$, $f, g \in C^\infty(M)$.

- $\mathbb{R}$ -linearity in second variable: For each $V \in \mathfrak{X}(M)$, the map $T \mapsto \nabla_V(T)$ is $\mathbb{R}$ -linear. This means that

$$\nabla_V(\lambda T + \mu S) = \lambda \nabla_V(T) + \mu \nabla_V(S)$$

- Product rule: The map ∇ satisfies the following product rule:

$$\nabla_V(fT) = V(f) \cdot T + f\nabla_V(T)$$

Definition 2.1.2 (Affine Connections) Let M be a smooth manifold. An affine connection of M is a connection

$$\nabla : \mathfrak{X}(M) \times \mathfrak{X}(M) \rightarrow \mathfrak{X}(M)$$

on the tangent bundle TM .

The directional derivative is the canonical affine connection on $\mathbb{R}^n$. In fact, every smooth manifold has such a canonical connection that generalizes the directional derivative.

Theorem 2.1.3 Every smooth manifold admits an affine connection.

2.2 Metric Connections

Definition 2.2.1 (Metric Connections) Let M be a smooth manifold. Let ∇ be an affine connection. We say that ∇ is a metric connection if the

$$\nabla(X, g(Y, Z)) = g(\nabla(X, Y), Z) + g(Y, \nabla(X, Z))$$

for all $X, Y, Z \in \mathfrak{X}(M)$.

2.3 The Levi-Civita Connection

Definition 2.3.1 (The Lie Bracket on Smooth Vector Fields) Let M be a smooth manifold. Let $X, Y \in \mathfrak{X}(M)$ be smooth vector fields. Define the Lie bracket of X and Y to be the vector field $[X, Y]$ given by the formula

$$[X, Y]_p(f) = X(Y_p(f)) - Y(X_p(f))$$

for $p \in M$ and $f \in C_{M,p}^\infty$.

Proposition 2.3.2 Let M be a smooth manifold. Let $X, Y \in \mathfrak{X}(M)$ be smooth vector fields. Let

(U, ϕ) be a chart on M . Let X and Y be given locally on the chart by the formula

$$X = \sum_{k=1}^n a_k \frac{\partial}{\partial x^k} \quad \text{and} \quad Y = \sum_{k=1}^n b_k \frac{\partial}{\partial x^k}$$

for $a_k, b_k : U \rightarrow \mathbb{R}$ smooth functions. Then the Lie bracket of X and Y is given locally on the chart by the formula

$$[X, Y] = \sum_{k=1}^n (X(b_k) - Y(a_k)) \frac{\partial}{\partial x^k} = \sum_{i=1}^n \left(\sum_{j=1}^n \left(a_j \frac{\partial b_i}{\partial x^j} - b_j \frac{\partial a_i}{\partial x^j} \right) \right) \frac{\partial}{\partial x^i}$$

Definition 2.3.3 (Levi-Civita Connections) Let (M, g) be a Riemannian manifold. Let $\nabla : \mathfrak{X}(M) \times \mathfrak{X}(M) \rightarrow \mathfrak{X}(M)$ be an affine connection on M . We say that ∇ is a Levi-Civita connection if the following are true.

- ∇ is a metric connection.
- For any $X, Y \in \mathfrak{X}(M)$, we have

$$\nabla(X, Y) - \nabla(Y, X) = [X, Y]$$

Theorem 2.3.4 (Existence and Uniqueness of Levi-Civita Connections) Let (M, g) be a Riemannian manifold. Then M has a unique Levi-Civita connection.

3 Taking Derivatives of the Vector Fields

3.1 Covariant Derivatives

Let M be a smooth manifold. Let $\gamma : I \rightarrow M$ be a smooth curve. Then $\text{im}(\gamma)$ is an embedded submanifold of M . Therefore it makes sense to talk about smooth vector fields on $\text{im}(\gamma)$. We overload the notation and denote the $\mathcal{C}^\infty(\text{im}(\gamma))$ -algebra of smooth vector spaces by

$$\mathfrak{X}(\gamma) = \{X : \text{im}(\gamma) \rightarrow TM \mid X \text{ is a smooth vector field on } \text{im}(\gamma)\}$$

Definition 3.1.1 (Covariant Derivatives Along a Curve) Let M be a smooth manifold. Let $\gamma : (a, b) \rightarrow M$ be a curve on M . Let $\nabla : \mathfrak{X}(M) \times \mathfrak{X}(M) \rightarrow \mathfrak{X}(M)$ be an affine connection on M . The covariant derivative of γ is a map

$$D_t : \mathfrak{X}(\gamma) \rightarrow \mathfrak{X}(\gamma)$$

such that

- $\mathbb{R}$ -linearity: $D_t(aV + bW) = aD_tV + bD_tW$ for $a, b \in \mathbb{R}$.
- Product rule: $D_t(fV) = f'V + fD_tV$ for $f \in C^\infty(a, b)$.
- Extendable: If $V \in \mathfrak{X}(\gamma)$ and there exists $\tilde{V} \in \mathfrak{X}(M)$ such that $\tilde{V}|_{\gamma(t)} = V(t)$ for all $t \in (a, b)$, then $D_tV = \nabla_{\gamma'(t)}\tilde{V}$.

Theorem 3.1.2 (Existence and Uniqueness of Covariant Derivatives) Let M be a smooth manifold. Let ∇ be an affine connection. For each smooth curve $\gamma : (a, b) \rightarrow M$, the connection ∇ determines a unique covariant derivative

$$D_t : \mathfrak{X}(\gamma) \rightarrow \mathfrak{X}(\gamma)$$

For each $t \in I$, choose a chart $(U, \phi = (x^1, \dots, x^n))$ for $\gamma(t) \in M$. Write V locally on the chart by

$$V_{\gamma(t)} = \sum_{i=1}^n a_i(\gamma(t)) \frac{\partial}{\partial x^i} \Big|_{\gamma(t)}$$

Then the unique associated covariant derivatives is given locally on the chart by the formula

$$(D_tV)_{\gamma(t_0)} = \sum_{i=1}^n \left(\frac{\partial a_i}{\partial x^i} \Big|_{\gamma(t_0)} \frac{\partial}{\partial x^i} \Big|_{\gamma(t_0)} + a_i(\gamma(t_0)) \nabla \left(\gamma'(t_0), \frac{\partial}{\partial x^i} \Big|_{\gamma(t_0)} \right) \right)$$

for $t_0 \in I$.

3.2 Parallel Transports

Definition 3.2.1 (Parallel Vector Fields along a Curve) Let M be a smooth manifold. Let $\gamma : I \rightarrow M$ be a curve. Let D_t be its associated covariant derivative. Let $X : M \rightarrow TM$ be a vector field. We say that X is parallel along γ if $D_tX = 0$.

Theorem 3.2.2 Let M be a smooth manifold. Let $\gamma : I \rightarrow M$ be a curve. Let ∇ be an affine connection. Let $t_0 \in I$ and $v_0 \in T_{\gamma(t_0)}M$. Then there exists a unique parallel vector field $V(t)$ along γ such that $V(t_0) = v_0$.

Definition 3.2.3 (Parallel Transports) Let M be a smooth manifold. Let $\gamma : (a, b) \rightarrow M$ be a curve on M . Let $t_0, t \in (a, b)$. The map

$$P_{t_0, t} : T_{\gamma(t_0)}(M) \rightarrow T_{\gamma(t)}(M)$$

defined by $v \mapsto X(t)$ where $X(t)$ is the unique parallel vector field along γ with $X(t_0) = v$.

4 Geometry on Riemannian Manifolds

4.1 Geodesics on Riemannian Manifolds

Let M be a smooth manifold. Let $\gamma : I \rightarrow M$ be a smooth curve on M . Then we can compute its differential to obtain a tangent vector

$$\gamma'(t) \in T_{\gamma(t)}M$$

for each $t \in I$. This defines a vector field in $\mathfrak{X}(\gamma)$.

Definition 4.1.1 (Geodesics) Let M be a smooth manifold. Let ∇ be an affine connection on M . A geodesic on M is a curve $\gamma : I \rightarrow M$ such that

$$D_t(\gamma'(t)) = 0$$

where D_t is the associated covariant derivative of ∇ and γ .

We have seen geodesics for metric spaces, but these are slightly different. We require geodesics on manifolds to only minimize the distance locally.

4.2 Minimizing Curves

Definition 4.2.1 (Minimizing Curves)

Let (M, g) be a Riemannian manifold. Let $\gamma : [a, b] \rightarrow M$ be a smooth curve in M . We say that γ is minimizing if

$$L(\gamma) = d_g(\gamma(a), \gamma(b))$$

Definition 4.2.2 (Locally Minimizing Curves)

Let (M, g) be a Riemannian manifold. Let $\gamma : [a, b] \rightarrow M$ be a smooth curve in M . We say that γ is locally minimizing if for all $t \in [a, b]$, there exists a neighbourhood $[c, d] \subseteq [a, b]$ of t such that $\gamma|_{[c, d]}$ is a minimizing curve.

Proposition 4.2.3

Let (M, g) be a Riemannian manifold. Let $\gamma : [a, b] \rightarrow M$ be a smooth curve in M . Then the following are true.

- If γ is a minimizing curve, then γ is a geodesic.
- If γ is a geodesic, then γ is a locally minimizing curve.

4.3 Geodesically Complete Riemannian Manifolds

Definition 4.3.1 (Geodesically Complete)

Proposition 4.3.2

Geodesically complete iff exp map well defined

Theorem 4.3.3 (Hopf-Rinow Theorem)

Let (M, g) be a connected Riemannian manifold. Then the following are equivalent.

- M is geodesically complete.
- M is complete as a metric space.
- Every closed and bounded set in M is compact.

Proposition 4.3.4

Let (M, g) be a connected Riemannian manifold. Let $p, q \in M$. Then there exists a minimizing geodesic from p to q .

5 Measuring the Curvature

5.1 The Curvature Tensor

Definition 5.1.1 (The Curvature Tensor)

Let (M, g) be a Riemannian manifold. Define the curvature tensor of M to be the map $R : \mathfrak{X}(M) \times \mathfrak{X}(M) \rightarrow \mathfrak{X}(M)$ given by

$$R(X, Y)Z = [\nabla_X, \nabla_Y]Z - \nabla_{[X, Y]}Z$$

Proposition 5.1.2

Let (M, g) be a Riemannian manifold. Then M is flat if and only if $R = 0$.

Recall: Flat iff locally isometric to $\mathbb{R}^n$.

5.2 The Second Fundamental Form

5.3 Gauss-Bonnet Theorem

Theorem 5.3.1 (The Gauss-Bonnet Formula) Let (M, g) be an oriented smooth 2-manifold. Let γ be a positively oriented curved polygon in M and let Ω be its interior. Then

$$\int_{\Omega} K dA + \int_{\gamma} \kappa_N ds + \sum_{i=1}^k \varepsilon_i = 2\pi$$

where

- K is the Gaussian curvature of g
- dA is the Riemannian volume form
- ε_i are the exterior angles of γ
- The second integral is taken with respect to arc length

Theorem 5.3.2 (Gauss-Bonnet Theorem) Let (M, g) be a smooth compact 2-dimensional Riemannian manifold. Let K be the Gaussian curvature of M and let k_g be the geodesic curvature of ∂M . Then

$$\int_M K dA + \int_{\partial M} k_g ds = 2\pi\chi(M)$$

Corollary 5.3.3 Let (M, g) be a smooth compact 2-dimensional Riemannian manifold without boundary. Let K be the Gaussian curvature of M . Then

$$\int_M K dA = 2\pi\chi(M)$$

6 Calculus on Manifolds

6.1 Integration of Functions on Riemannian Manifolds

Definition 6.1.1 (The Riemannian Volume Form)

Let (M, g) be an orientable Riemannian manifold. Define the volume form dV of M to be given locally at $p \in M$ on the chart $(U, x^1, \dots, x^n)$ by

$$(dV)_p = \sqrt{\det(g_{i,j}(p))} (dx^1)_p \wedge \dots \wedge (dx^n)_p$$

where g is given locally at the chart by $g_p = \sum_{1 \leq i, j \leq n} g_{i,j}(p) (dx^i)_p \otimes (dx^j)_p$.

Let (M, g) be a compact orientable Riemannian manifold. Let $f : M \rightarrow \mathbb{R}$ be a compactly supported smooth function. Then we may define the integral of f over M to be the value $\int_M f dV$.

Definition 6.1.2 (The Volume of a Compact Orientable Manifold)

Let (M, g) be a compact orientable Riemannian manifold. Define the volume of M to be

$$\text{Vol}(M) = \int_M 1 dV$$

6.2 Differential Operators

Definition 6.2.1 (The Gradient of a Function)

Let (M, g) be a Riemannian manifold. Let $f : M \rightarrow \mathbb{R}$ be a smooth function. Define the gradient of f to be

$$\text{grad}(f) = (df)^\#$$

By definition, the gradient of f is the unique section of TM such that

$$(df)_p(w) = g_p(\text{grad}(f)_p, w)$$

for all $p \in M$ and $w \in T_p M$.

Proposition 6.2.2

Let (M, g) be a Riemannian manifold. Let $(U, x^1, \dots, x^n)$ be a chart in M . Let $f : M \rightarrow \mathbb{R}$ be a smooth function. Let g be given locally at the chart by

$$g_p = \sum_{1 \leq i, j \leq n} g_{i,j}(p) (dx^i)_p \otimes (dx^j)_p$$

Let $(g^{i,j}(p)) = (g_{i,j}(p))^{-1}$ be the inverse of the matrix. Then locally at U , the gradient of f at p is given by

$$\text{grad}(f)(p) = \sum_{1 \leq i, j \leq n} \frac{\partial f}{\partial x^i} \Big|_p g^{i,j}(p) \frac{\partial}{\partial x^j} \Big|_p$$

Example 6.2.3

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a smooth function. Then the gradient of f at $p \in \mathbb{R}^n$ is given by

$$\text{grad}(f)(p) = \sum_{i=1}^n \frac{\partial f}{\partial x_i} \Big|_p \frac{\partial}{\partial x_i} \Big|_p$$

Definition 6.2.4 (The Divergence of a Vector Field)

Let (M, g) be a Riemannian manifold. Let $X : M \rightarrow TM$ be a smooth vector field. Define the divergence of X to be the unique smooth function $\text{div}(X) : M \rightarrow \mathbb{R}$ such that for each $p \in M$ and

$(U, x^1, \dots, x^n)$ a chart at p , we have

$$(d(X \lrcorner dV))_p = (\operatorname{div}(X)(p))(dV)_p$$

where dV is the local top form given by $(dV)_p = \sqrt{\det(g_{i,j}(p))}(dx^1)_p \wedge \dots \wedge (dx^n)_p$ and g is given locally at the chart by $g_p = \sum_{1 \leq i,j \leq n} g_{i,j}(p)(dx^i)_p \otimes (dx^j)_p$.

Proposition 6.2.5

Let (M, g) be a Riemannian manifold. Let $(U, x^1, \dots, x^n)$ be a chart in M . Let $X : M \rightarrow TM$ be a smooth vector field. Suppose that g and X are given locally at the chart by

$$g_p = \sum_{1 \leq i,j \leq n} g_{i,j}(p)(dx^i)_p \otimes (dx^j)_p \quad \text{and} \quad X(p) = \sum_{i=1}^n a_i \frac{\partial}{\partial x^i} \Big|_p$$

respectively. Then locally at U , the divergence of X at $p \in U$ is given by

$$\operatorname{div}(X)(p) = \frac{1}{\sqrt{\det(g_{i,j}(p))}} \sum_{k=1}^n \frac{\partial(a_k \circ \sqrt{\det(g_{i,j}))})}{\partial x^k} \Big|_p$$

Example 6.2.6

Let $X : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a smooth vector field whose components are given by $X_i : \mathbb{R}^n \rightarrow \mathbb{R}$. Then the divergence of X at $p \in \mathbb{R}^n$ is given by

$$\operatorname{div}(X)(p) = \sum_{i=1}^n \frac{\partial X_i}{\partial x_i} \Big|_p$$

Definition 6.2.7 (The Laplacian of a Function)

Let (M, g) be a Riemannian manifold. Let $f : M \rightarrow \mathbb{R}$ be a smooth function. Define the Laplacian of f to be the function $\Delta f : M \rightarrow \mathbb{R}$ given by

$$\Delta f = \operatorname{div}(\operatorname{grad}(f))$$

Proposition 6.2.8

Let (M, g) be a Riemannian manifold. Let $f : M \rightarrow \mathbb{R}$ be a smooth function. Let $(U, x^1, \dots, x^n)$ be a chart in M . Let g be given locally at the chart by

$$g_p = \sum_{1 \leq i,j \leq n} g_{i,j}(p)(dx^i)_p \otimes (dx^j)_p$$

Let $(g^{i,j}(p)) = (g_{i,j}(p))^{-1}$ be the inverse of the matrix. Then locally at U , the laplcian of f at p is given by

$$\Delta(f)(p) = \frac{1}{\sqrt{\det(g_{i,j}(p))}} \sum_{1 \leq k,l \leq n} \frac{\partial(g^{k,l} \circ \sqrt{\det(g_{i,j}))} \circ \frac{\partial f}{\partial x^l})}{\partial x^k} \Big|_p$$

Example 6.2.9

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a smooth function. Then the laplacian of f at $p \in \mathbb{R}^n$ is given by

$$\Delta(f)(p) = \sum_{i=1}^n \frac{\partial^2 f}{\partial x_i^2} \Big|_p$$

6.3 Curl and The Case of Dimension 3

Definition 6.3.1 (The β -Operator)

Let (M, g) be an oriented Riemannian manifold of dimension 3. Define the β -operator to be the map $\beta : TM \rightarrow \bigwedge_{i=1}^2(T^*M)$ by

$$\beta(X) = X \lrcorner dV$$

Lemma 6.3.2

Let (M, g) be an oriented Riemannian manifold of dimension 3. Then β is a smooth bundle isomorphism.

Definition 6.3.3 (The Curl of a Vector Field)

Let (M, g) be an oriented Riemannian manifold of dimension 3. Let $X : M \rightarrow TM$ be a smooth vector field. Define the curl of X to be the vector field $\text{curl}(X) : M \rightarrow TM$ given by

$$\text{curl}(X) = \beta^{-1}(d(X^\flat))$$

Proposition 6.3.4

Let (M, g) be an oriented Riemannian manifold of dimension 3. Then the following diagram is commutative:

$$\begin{array}{ccccccc} \mathcal{C}^\infty(M) & \xrightarrow{\text{grad}} & \mathfrak{X}(M) & \xrightarrow{\text{curl}} & \mathfrak{X}(M) & \xrightarrow{\text{div}} & \mathcal{C}^\infty(M) \\ \text{id} \downarrow & & \downarrow \flat & & \downarrow \beta & & \downarrow \flat \\ \Omega^0(M) & \xrightarrow{d} & \Omega^1(M) & \xrightarrow{d} & \Omega^2(M) & \xrightarrow{d} & \Omega^3(M) \end{array}$$

6.4 Calculus on Manifolds

Let M be an oriented smooth manifold of dimension n . let $p \in \partial M$. Then $T_p(\partial M)$ is an $(n-1)$ -dimensional vector subspace of $T_p(M)$. Recall that a boundary chart of M at $p \in M$ is a chart (U, ϕ) such that the following are true.

- $\phi(U) \subseteq \{(x_1, \dots, x_n) \in \mathbb{R}^n \mid x_n \geq 0\}$.
- $\phi^{-1}(\{(x_1, \dots, x_n) \in \mathbb{R}^n \mid x_n = 0\}) \subseteq \partial M$.
- $\phi(p) = 0$.

Definition 6.4.1 (Unit Normal Vectors)

Let (M, g) be a Riemannian manifold of dimension n with boundary. Let $p \in \partial M$. Define the normal vector $n(p)$ of p to be the unique vector in $T_p(M)$ such that the following are true.

- $n(p)$ is normal to the hyperplane $T_p(\partial M)$ in $T_p(M)$ with respect to the inner product g_p .
- $n(p)$ is a unit vector.
- Choose a boundary chart (U, ϕ) at p . Denote $(\phi_*)_p : T_p(M) \rightarrow T_{\phi(p)}(\mathbb{R}^n)$ the differential of ϕ . Then n th coordinate of $(\phi_*)_p(n_p)$ is less than 0 under the standard basis of $T_{\phi(p)}(\mathbb{R}^n)$.

Lemma 6.4.2

Let (M, g) be a Riemannian manifold of dimension n with boundary. If M is orientable, then the assignment $p \mapsto n(p)$ is a smooth map from ∂M to TM .

Theorem 6.4.3 (The Gradient Theorem)

Let $M \subseteq \mathbb{R}^n$ be a smooth compact connected 1-dimensional manifold. Let $r : [a, b] \rightarrow M$ be a smooth curve such that $\text{im}(r) = M$. Let $f \in \mathcal{C}^\infty(M)$ be a smooth function. Then we have

$$\int_M \nabla f = f(r(b)) - f(r(a))$$

Theorem 6.4.4 (The Divergence Theorem)

Let $M \subseteq \mathbb{R}^n$ be a smooth compact manifold of dimension n with boundary. Let $F : M \rightarrow \mathbb{R}^n$ be a smooth function. Then we have

$$\int_M \operatorname{div}(F) = \int_{\partial M} F \cdot n(x)$$

where both integrals are Riemann integrals.

Theorem 6.4.5

7 The Hodge Star Operator

7.1 The Hodge Star Operator

Definition 7.1.1 (The Hodge Star Operator)

Let (M, g) be an orientable Riemannian manifold. Let $n = \dim(M)$.

- Let $p \in M$. Define $*_p : \Lambda^k T_p^*(M) \rightarrow \Lambda^{n-k} T_p^*(M)$ to be the unique $\mathbb{C}$ -linear isomorphism such that

$$\alpha \wedge (*_p \beta) = g_p(\alpha, \beta)(dV)_p$$

where $\alpha, \beta \in \Lambda^k T_p^*(M)$ and Vol is the volume form of M associated to g , and $\langle -, - \rangle_p$ is the induced metric on $\Lambda^k(T_p^*(M))$.

- Define the Hodge star operator of M to be the map

$$* \in \Gamma(M, \text{Hom}(\Omega^k(M), \Omega^{n-k}(M)))$$

given by $p \mapsto *_p$.

Lemma 7.1.2

Let (M, g) be an orientable Riemannian manifold. Then the following are true regarding the Hodge star operator.

- $*f = f dV$ for any smooth function $f \in C^\infty(M)$.
- $**\omega = (-1)^{k(n-k)}\omega$ for any smooth k -form $\omega \in \Omega^k(M)$.

Lemma 7.1.3

Let (M, g) be an orientable Riemannian manifold. Let $X : M \rightarrow TM$ be a smooth vector field. Then the following are true.

- $X \lrcorner dV = *(X^\flat)$.
- $\text{div}(X) = (* \circ d \circ *)(X^\flat)$

Lemma 7.1.4

Let (M, g) be an orientable Riemannian manifold of dimension 3. Let $X : M \rightarrow TM$ be a smooth vector field. Then we have

$$\text{curl}(X) = (* \circ d(X^\flat))^\#$$

7.2 The L^2 -Norm

Definition 7.2.1 (The L^2 -Norm)

Let (M, g) be an orientable Riemannian manifold. Define the L^2 -norm on M to be the map $\langle -, - \rangle_{L^2} : \Omega^k(M) \times \Omega^k(M) \rightarrow \mathbb{R}$ by the formula

$$\langle \alpha, \beta \rangle_{L^2} = \int_M \langle \alpha, \beta \rangle_g dV$$

where $\langle -, - \rangle_g$ is the induced inner product on k -forms.

Lemma 7.2.2

Let (M, g) be an orientable Riemannian manifold. Then $\langle -, - \rangle_{L^2}$ defines an inner product on the compactly supported smooth k -forms on M for any $k \in \mathbb{N}$.

Lemma 7.2.3

Let (M, g) be an orientable Riemannian manifold. Let $\alpha, \beta \in \Omega^k(M)$ be compactly supported smooth k -forms. Then we have

$$\langle \alpha, \beta \rangle_{L^2} = \int_M \alpha \wedge * \beta$$

7.3 The Adjoint Differential Operator

Definition 7.3.1 (The Adjoint Differential Operator)

Let (M, g) be an orientable Riemannian manifold. Define the operator $d^* : \Omega^k(M) \rightarrow \Omega^{k-1}(M)$ by

$$d^*(\omega) = (-1)^{n(k+1)+1}(* \circ d \circ *)(\omega)$$

for $\omega \in \Omega^k(M)$. Also define $d^*(\omega) = 0$ for $\omega \in \Omega^0(M)$ by convention.

Lemma 7.3.2

Let (M, g) be an orientable Riemannian manifold. Then we have $d^* \circ d^* = 0$.

Proposition 7.3.3

Let (M, g) be a compact orientable Riemannian manifold. Then we have

$$\langle d^*\omega, \eta \rangle_{L^2} = \langle \omega, d\eta \rangle_{L^2}$$

for any $\omega \in \Omega^{k+1}(M)$ and $\eta \in \Omega^k(M)$.

Definition 7.3.4 (The Laplace-Beltrami Operator)

Let M be an orientable Riemannian manifold. Define the Laplace-Beltrami operator on M to be the map $\Delta : \Omega^k(M) \rightarrow \Omega^k(M)$ given by

$$\Delta = d^* \circ d + d \circ d^*$$

Lemma 7.3.5

Let M be an orientable Riemannian manifold. Then the following are true regarding the Laplace-Beltrami operator.

- $* \circ \Delta = \Delta \circ *$,
- $\langle \Delta\alpha, \beta \rangle_{L^2} = \langle \alpha, \Delta\beta \rangle_{L^2}$ for any compactly supported differential forms $\alpha, \beta \in \Omega^k(M)$.

7.4 The Hodge Theorem for Riemannian Manifolds

Definition 7.4.1 (Harmonic Forms)

Let M be an orientable Riemannian manifold. Let $\alpha \in \Omega^k(M)$. We say that α is harmonic if $\alpha \in \ker(\Delta)$.

Lemma 7.4.2

Let M be an orientable Riemannian manifold. Let $\alpha \in \Omega^k(M)$. Then α is harmonic if and only if $d(\alpha) = 0$ and $d^*(\alpha) = 0$.

Definition 7.4.3 (The Space of Harmonic Forms)

Let M be an orientable Riemannian manifold. Define the space of harmonic k -forms to be

$$\mathcal{H}^k(M) = \ker(\Delta)$$

Theorem 7.4.4 (Hodge Theorem)

Let M be a compact orientable Riemannian manifold. Then the following are true.

- $\mathcal{H}^k$ is finite dimensional for all $k \in \mathbb{N}$.
- There is a decomposition

$$\Omega^k(M) \cong \mathcal{H}^k(M) \oplus \text{im}(\Delta)$$

Corollary 7.4.5

Let M be an orientable Riemannian manifold. Then the quotient map induces an isomorphism

$$\mathcal{H}^k(M) \cong H_{\text{dR}}^k(M)$$

for all $k \in \mathbb{N}$.